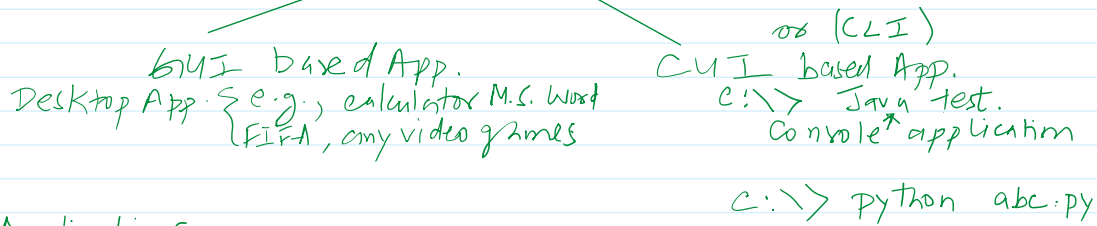


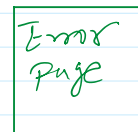
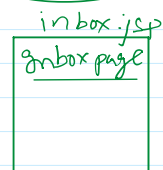
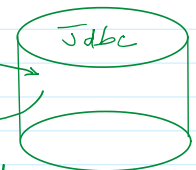
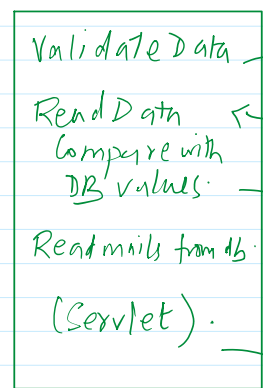
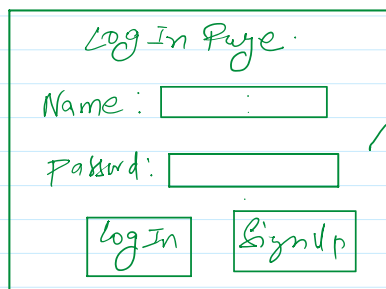
Standalone applications



Web Applications.

gmail
instagram.
github.

- * JDBC
- * Servlets.
- * JSP



error.jsp

View Components -

{ presentation logic }

or JSP (Java Server Pages)

Editions of Java

1. Java Standard Edition. (JDK / JSE / J2SE)

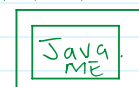
⇒ Core Java + JDBC

② Java Enterprise Edition (J2EE / JEE)

[Servlets + JSP].

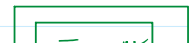
③ Java Micro Edition (J2ME / JME).

Android + SDK.



only for
Android

Kotlin.



cross
platform

New Java Software's:



The world's most popular modern development platform

The world's most popular modern development platform, Java SE is the programming language of choice for enterprise applications. Java SE reduces costs, shortens development time, drives innovation, and

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- OCI SDK for Java: The Oracle Cloud Infrastructure SDK for Java enables you to write code to manage OCI resources.
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- Simplified Java Management Service gives you insights into all your Java deployments, on OCI and on-premises.
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Java SE

Java on OCI

Oracle WebLogic Server

Java Card

Java Management Service

The world's leading platform for developing and running enterprise Java applications

Oracle WebLogic Server offers a robust, mature, and scalable implementation of Oracle Java Enterprise Edition (EE) and Jakarta EE to run enterprise Java applications in the cloud or on-premises.

[Explore Oracle WebLogic Server](#)

Products

- Oracle WebLogic Server
- Oracle WebLogic Server for Oracle Cloud Infrastructure
- Oracle Coherence
- Helidon



Java SE

Java on OCI

Oracle WebLogic Server

Java Card

Java Management Service

Improves security for Internet of Things (IoT) edge devices

Java Card enables secure elements, such as smart cards and other tamper-resistant security chips, to host applications based on Java technology. Store and update multiple applications on a single, resource-constrained device.

Explore Java Card

Documentation

Features

- Interoperable
- Multiapplication, multitenant
- Secure execution environment
- Extensible and updatable



Java SE

Java on OCI

Oracle WebLogic Server

Java Card

Java Management Service

Java Management Service

Enterprises often run multiple versions of Java across desktops, laptops, servers, and cloud environments, making it challenging to maintain security and compliance. Outdated Java versions can expose organizations to vulnerabilities. Java Management Service (JMS) is an Oracle Cloud Infrastructure (OCI) service that provides centralized, real-time monitoring and management of Java environments both on-premises and in the cloud. JMS gives you a single, unified console to discover, track, and manage all Java runtimes—whether they are Oracle or non-Oracle, client or server—across your organization. JMS helps mitigate security risks by quickly identifying outdated Java installations and surfacing vulnerabilities in third-party libraries. JMS empowers teams to optimize performance, accelerate modernization, and maintain compliance across the entire Java estate.

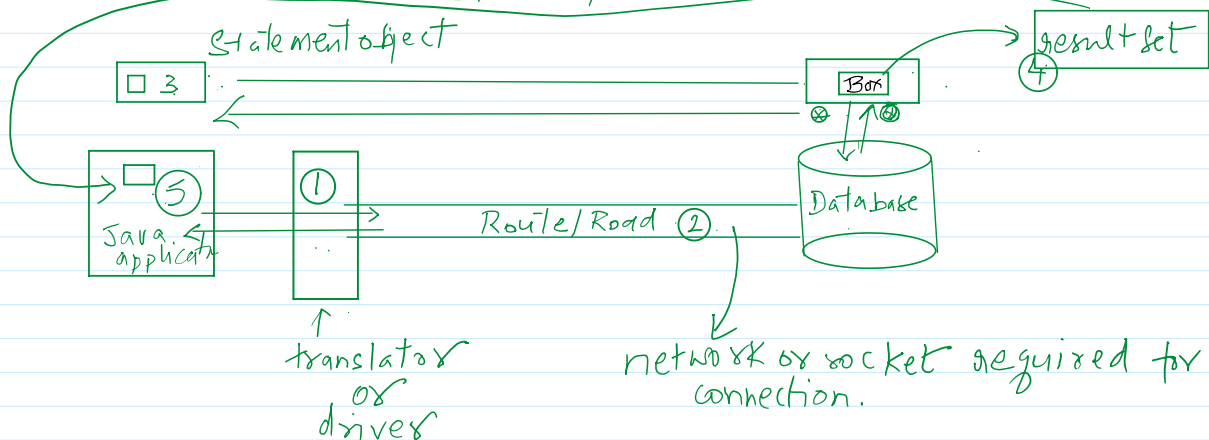
Basic JMS capabilities, including Java runtime discovery and usage tracking, are available free to all users, while advanced features, such as migration analysis, performance monitoring, vulnerability scanning, and remote management, are included for Oracle Java SE Subscribers or customers running Java workloads on OCI.

Learn more about JMS

JDBC 4.3V

Servlets 6.0V

JSP 4.0V



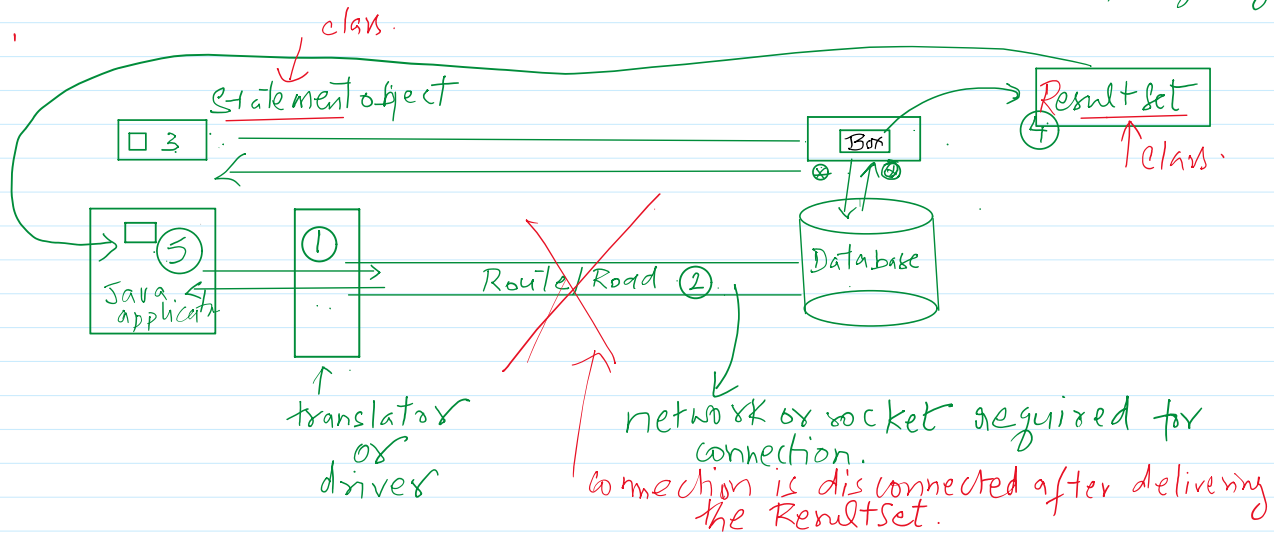
What is the purpose of statement object?

Basic component of JDBC Application:

1. Database Driver: Translator to convert Java specified calls to database specific calls or database specific calls to Java.
2. Network socket or connection: - (Road) or (Route).

3. Statement object (Vehicle) : to send our SQL query to database and brings back ResultSet.

4. ResultSet (box) : In which database returns a results of a query



Steps To Develop JDBC Application (Standard Steps).

db (database)

1. Load and Register driver
2. Establish connection b/w Java app. & db
3. Creation of Statement object (Create statement object).
4. Send and execute SQL query.
5. Process results from ResultSet.
6. Close connection.

```
import java.sql.*;
```

```
class JDBC {
```

```
    public static void main (String[] args) {
```

```
        ① → Class.forName("oracle.jdbc.OracleDriver");
```

```
        ② → Connection con = DriverManager.getConnection(url, user, pwd);
```

```
        ③ → Statement st = con.createStatement();
```

```
        ④ → ResultSet rs = st.executeQuery("select * from emp.");
```

```
        ⑤ → while (rs.next()) {
```

```
            System.out.println("id" + rs.getInt(1) + "... " + rs.getString(2) +
```

load the driver
on the
memory.

✓ → write (0.0, mexli) /

```
S.o.print("In" + rs.getInt(1) + "... " + rs.getString(2) +
        "... " + rs.getDouble(3) + "... " + rs.getString(4));
```

```

        }
    }
}

```

What is job?

It is a technology, which is used to communicate with database.

jdbc is a part of java standard edition. (J2SE/JSE)



JDBC Specification is defined by Java vendor (Oracle).

Who is the implementor of JDBC specification? Database vendor

Database vendor provided implementation is called Driver software.

JDBC Features

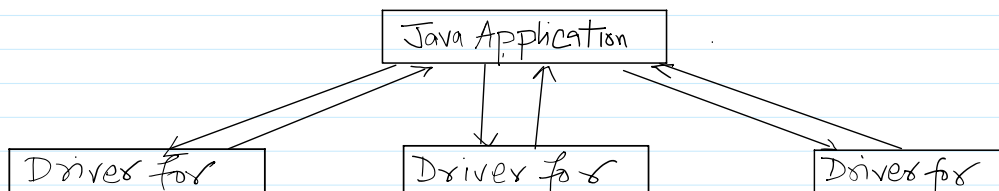
1. Standard API - Database Independent API.
2. Java - Platform Independent technology

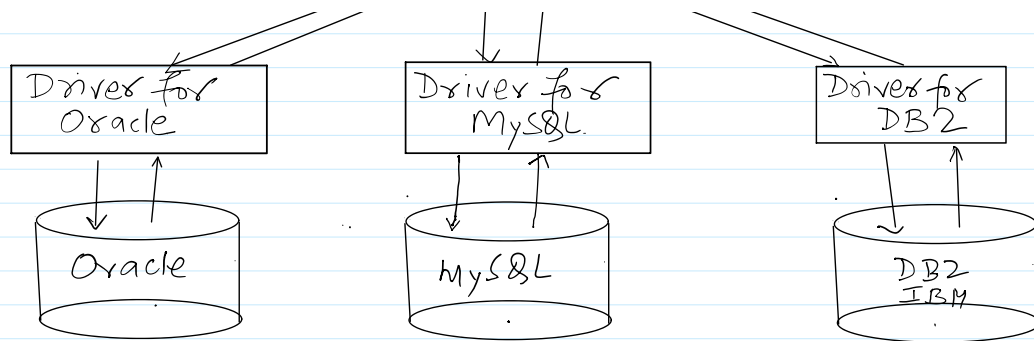
3. **CRUD**

CRUD	Meaning
C	Create
R	Retrieve
U	Update
D	Delete

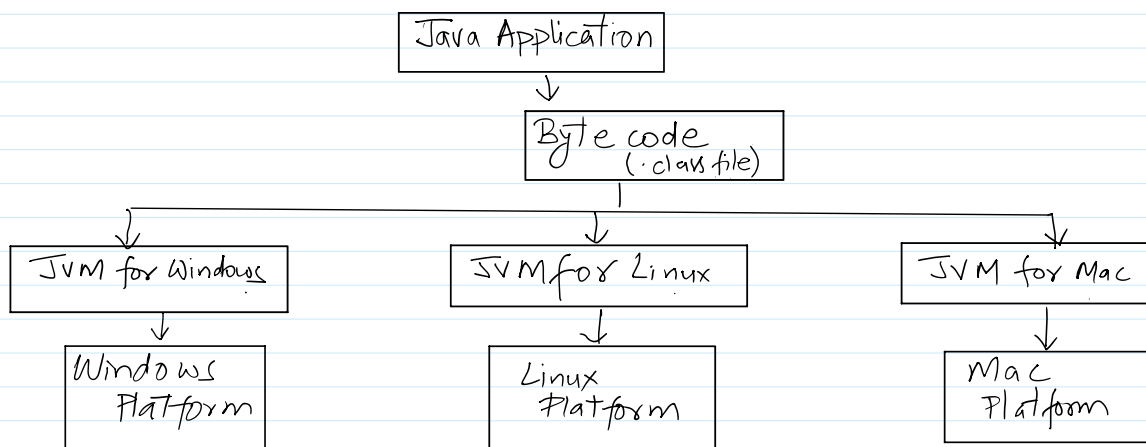
4. Huge vendor support.

How? Java is Platform independent & Database independent



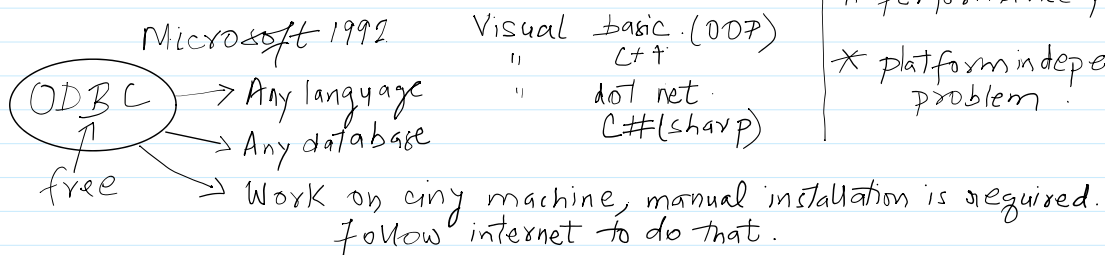


Java is database independent
Driver is database dependent.

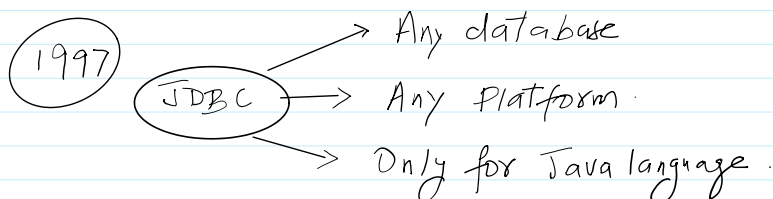
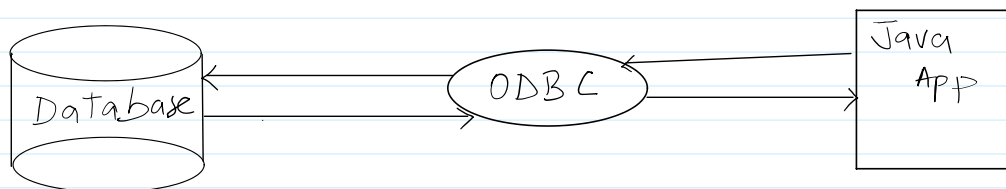


Java is platform independent because of **JVM**

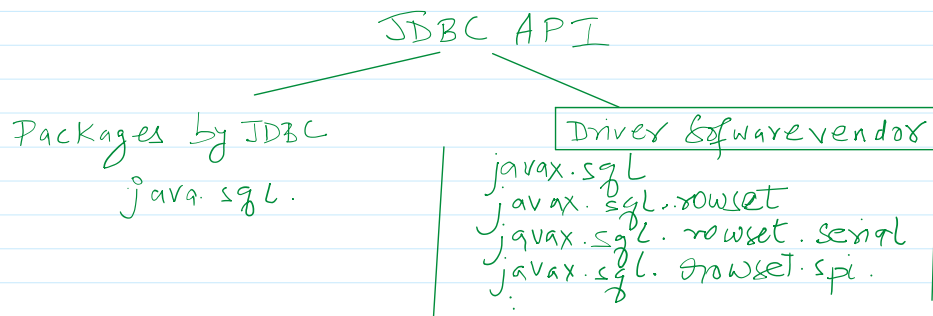
But JVM is platform dependent.



- * performance problems
- * platform indepenence problem.



ODBC	JDBC
ODBC stands for Open Database Connectivity.	JDBC stands for Java Database Connectivity.
Introduced by Microsoft in 1992.	Introduced by Sun Microsystem in 1997.
We can use ODBC for any language like C, C++, Java etc.	So we can use JDBC only for Java programming language.
Mostly ODBC drivers are developed in native languages like C or C++.	Mostly JDBC drivers are developed in Java.
For Java application it is not recommended to use ODBC because performance will be down due to internal conversions and application will become platform dependent.	For Java application it is highly recommended to use JDBC because there are no performance and platform dependent problems.



- * JDBC API provides several classes and interfaces.
- * Programmer can use these classes and interface to communicate with database.
- * Driver software vendor can use JDBC API while developing driver software.
- * JDBC API contains 2 packages.

1. `java.sql` package.

Contains basic classes and interfaces which can be used for database communication.

Interfaces

Driver
Connection
Statement
PreparedStatement
CallableStatement
ResultSet

Classes

DriverManager
Date
Time
Timestamp
Types

2. `javax.sql` package.

It contains more advance classes and interfaces which can be used for database communication.

Contains multiple sub-packages.

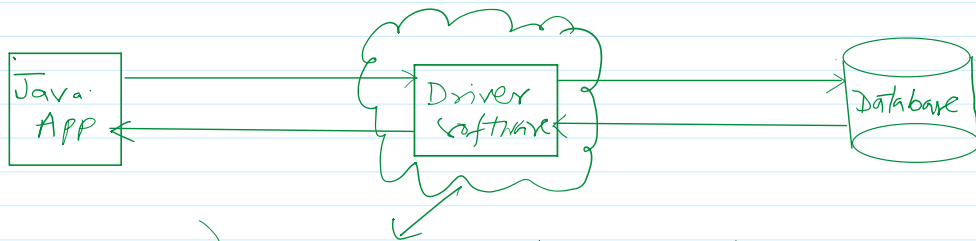
`javax.sql.rowset`
`javax.sql.rowset.serial`
`javax.sql.rowset.spi`

Interfaces

DataSource
Rowset
RowsetListener
ConnectionEventListener

Classes

RowsetEvent
ConnectionEvent

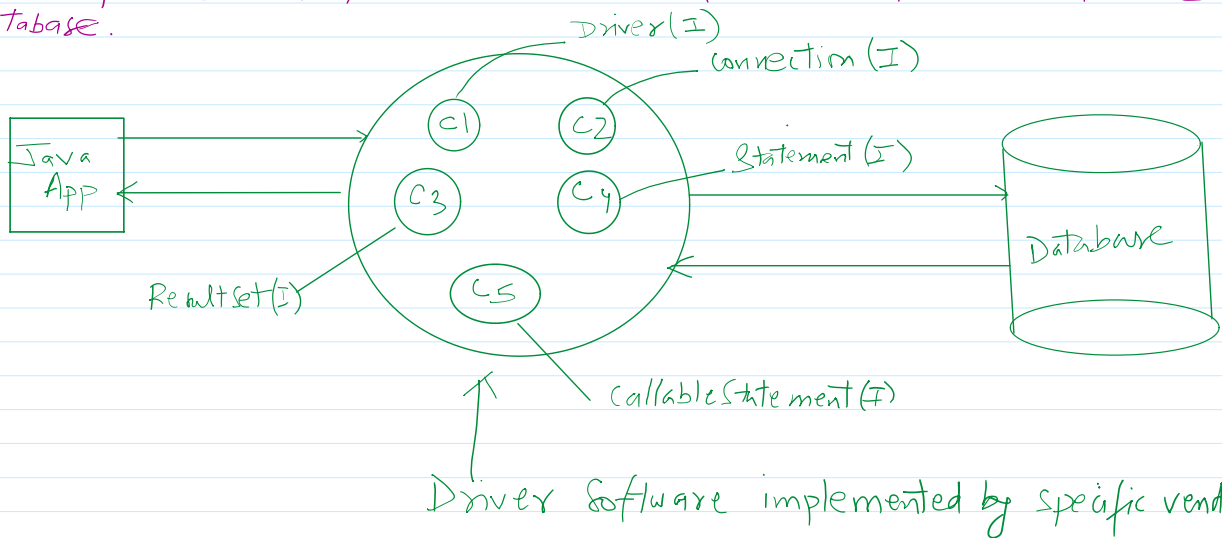


(A group of class) collections of implementation classes of JDBC API.

Oracle Driver Software is a collection of implementation classes of JDBC API interfaces.

Application.

- (*) Programmers are not responsible to provide implementation of JDBC API interfaces.
- (*) Most of the times Database vendor is responsible to provide implementation as the part of Driver software.
- (*) Every Driver software is a collection of classes implementing various interfaces of JDBC API, which can be used to communicate with a particular database.
- (*) For example, Driver Software for Oracle means collection of implementation classes of JDBC API, which can be used to communicate with Oracle database.



SUN JDK.

Type-1

`sun.jdbc.odbc.JdbcOdbcDriver`

Type-4 (Thin)

`oracle.jdbc.OracleDriver`

- (*) Every Driver Software is identified with a special class which is nothing but Driver class.
- (*) It is the implementation class of `java.sql.Driver` interface

communicating with database drivers.

(*) It is the implementation class of `java.sql.Driver` interface

```
public class OracleDriver implements Driver {
```

```
    ...
```

```
}
```

eg1 Type-1 Driver software implemented by SUN MICRO SYSTEM as the part of jdk. The corresponding driver class name is.

sun.jdbc.odbc.JdbcOdbcDriver.

↑
package name

↑
class Name.

eg2- Thin driver provided by Oracle Corporation & corresponding driver class name is.

oracle.jdbc.OracleDriver

↑
package name

↑
class name

→ free of cost.

Types of JDBC Drivers.

1. Type-1 Driver (JDBC- ODBC Bridge Driver or Bridge driver).
2. Type-2 Driver (Partly Java or Native-API - Partly Java driver or Native driver).
3. Type-3 Driver (All Java Net protocol Drivers or Network Protocol Driver or Middleware driver).
4. Type-4 Driver (Pure Java Drivers, (or) Native Protocol driver (or) Thin Driver.)
5. Type-5 Driver (Progress Direct). (Not free, Proprietary software)

(*) While communicating with database, we have to convert Java call into database specific calls and database specific calls into Java calls. For this Driver software is required.

(*) In the market thousands of Drivers are available. But based on functionality and architecture all drivers are divided into 4 types